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Automation project for news portal using Online Khabar as the primary system under test with extension to an e-commerce site.

0.System Overview

This test plan defines the strategy, objectives, scope, resources and schedule for automating the e-commerce website.

- OnlineKhabar is a high-traffic digital news platform with millions of visits and real-time updates.
- Features:
 - Category-bases navigation (Business, Tech, etc.)
 - Dynamic content (ads, live updates)
 - Heavy traffic + frequent DOM changes
- Challenges:
 - Dynamic ads & popups
 - Lazy loading
 - High DOM complexity

1.Master Test Plan (REALISTIC + INDUSTRY LEVEL)

Objective

Automate:

- Functional workflows
- UI validation
- Dynamic content behavior
- Performance & scalability

Scope

In Scope

- Homepage
- Category navigation
- Article reading
- Search
- Ads and popups
- Load performance

Out of Scope

- Backend APIs
- CMS publishing system

Test Strategy

Layer	Tool
UI Automation	Selenium
Framework	Pytest
Design Pattern	POM
Performance	JMeter
Reporting	Html

Test Scenarios

Functional

- Load homepage
- Navigate category
- Open article
- Scroll page
- Search keyword

UI

- Ads do not overlap content
- Images visible
- Mobile responsiveness

Non-Functional

- Load time < 3 sec
- Stable under 100 users

2.Automation Framework (PYTEST + POM)

Structure

Automation_projects/

|---- tests/

|----pages/

|----locators/

|----reports/

|----conftest.py

Web Elements Covered

- Text fields
- Buttons
- Dropdowns
- Links
- Dynamic elements (AJAX content)

Handling Dynamic Content

- Use explicit waits (WebDriverWait)
- Identify elements using dynamic XPATH
- Handle AJAX loading

Example:

```
WebDriverWait(driver, 10).until(  
    EC.visibility_of_element_located((By.ID, "news"))  
)
```

Performing Actions

- Click actions
- Form submission
- Drag and drop (if applicable)
- Navigation between pages

3. Prerequisites

3.1 Software Requirements

- Programming language (Python, JavaScript)
- Selenium WebDriver
- Browser drivers:
 - ChromeDriver (for Chrome)
 - GeckoDriver (for Firefox)
- Test framework (Pytest+POM, JMeter)

3.2 Environment Setup

- Install browsers: Chrome, Firefox
- Set up environment variables for driver
- Ensure compatible driver versions with browsers

4. Test Environment

4.1 Browser:

- Google Chrome (latest, previous)
- Mozilla Edge (latest, previous)
- Microsoft Edge (latest)

4.2 Operating Systems:

- Windows 11

4.3 Tools:

- Selenium WebDriver
- Pytest (POM)
- WebDriver Manager

5. Tools and Technologies

Category	Tool
Language	Python
Automation	Selenium WebDriver
Testing Framework	Pytest (POM)
IDE	PyCharm
Reporting	PyTest HTML
Parallel Execution	PyTest-xdist

6. Performance Testing Scenario

Scenario:

Simulate 100 concurrent users browsing system and search for related lists for information.

Metrics:

- Response time
- Throughput
- Error rate

Tools:

- JMeter (for load testing)

Report: PerformanceTesting_for Intern_Task.pdf

Screenshots:

Load Testing:

Summary Report

Name:

Comments:

· Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	101	17364	1206	28368	5214.78	0.00%	46.7/min	328.57	0.11	431912.6
TOTAL	101	17364	1206	28368	5214.78	0.00%	46.7/min	328.57	0.11	431912.6

View Results Tree

Name:

Comments:

· Write results to file / Read from file

Filename

Search: ☐ Case sensitive ☐ Regular exp.

Text

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

HTTP Request

Sampler result

Request

Response data

Thread Name:users 1-1

Sample Start:2026-05-07 07:12:01 PDT

Load time:1206

Connect Time:933

Latency:984

Size in bytes:431915

Sent bytes:142

Headers size in bytes:1065

Body size in bytes:430850

Sample Count:1

Error Count:0

Data type ("text"|"bin"|""):text

Response code:200

Response message:OK

HTTPSampleResult fields:

ContentType: text/html; charset=UTF-8

DataEncoding: UTF-8

5

View Results in Table

Name:

Comments:

· Write results to file / Read from file

Filename: Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	SampleTime(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	07:12:01.084	users 1-1	HTTP Request	1206	✓	431915	142	984	933
2	07:13:41.771	users 1-4	HTTP Request	6613	✓	431911	142	190	28
3	07:13:41.836	users 1-6	HTTP Request	7416	✓	431921	142	469	215
4	07:13:41.979	users 1-11	HTTP Request	7546	✓	431911	142	794	329
5	07:13:41.681	users 1-1	HTTP Request	8394	✓	431911	142	241	52
6	07:13:41.711	users 1-2	HTTP Request	8798	✓	431913	142	241	37
7	07:13:41.740	users 1-3	HTTP Request	8805	✓	431913	142	201	24
8	07:13:41.859	users 1-7	HTTP Request	8974	✓	431917	142	483	206
9	07:13:41.836	users 1-5	HTTP Request	9202	✓	431917	142	509	229
10	07:13:42.009	users 1-12	HTTP Request	9961	✓	431913	142	766	297
11	07:13:42.160	users 1-17	HTTP Request	11098	✓	431910	142	923	299
12	07:13:43.389	users 1-58	HTTP Request	11392	✓	431906	142	3070	1952
13	07:13:43.898	users 1-75	HTTP Request	12021	✓	431906	142	3300	1848
14	07:13:44.021	users 1-79	HTTP Request	12086	✓	431906	142	3214	1739
15	07:13:44.078	users 1-81	HTTP Request	12030	✓	431904	142	3157	1686
16	07:13:43.928	users 1-76	HTTP Request	12185	✓	431908	142	3328	1832
17	07:13:43.960	users 1-77	HTTP Request	12167	✓	431908	142	3321	1805
18	07:13:44.230	users 1-86	HTTP Request	11900	✓	431906	142	3315	1565
19	07:13:44.141	users 1-83	HTTP Request	11996	✓	431910	142	3404	1654
20	07:13:43.868	users 1-74	HTTP Request	12516	✓	431906	142	3347	1877
21	07:13:44.109	users 1-82	HTTP Request	12499	✓	431912	142	3401	1685
22	07:13:44.318	users 1-89	HTTP Request	12537	✓	431914	142	3474	1499
23	07:13:44.379	users 1-91	HTTP Request	12476	✓	431908	142	3421	1438
24	07:13:44.260	users 1-87	HTTP Request	12664	✓	431910	142	3597	1557
25	07:13:41.890	users 1-8	HTTP Request	15039	✓	431911	142	424	168

☐ Scroll automatically? ☐ Child samples?

No. of Samples: 101

Latest Sample: 20160

Average: 17364

Deviation: 5214

Scenario 1: Homepage Load Test

Parameter	Value
Users	100
Ramp Up	30 sec
Duration	5 min
Expected Avg Response	< 3 sec

Scenario 2: Search Load Test

Parameter	Value
Concurrent Users	50
Search Keyword	Nepal
Expected Response	< 2 sec

Scenario 3: Stress Test

Parameter	Value
Users	500
Objective	Server stability
Expected	No crash

7. Web Elements Used in Automation

The automation uses various web elements.

- Input Fields
- Buttons
- Links
- Dropdowns
- Alerts

- Dynamic Tables
- Checkboxes
- Radio Buttons

Handling Dynamic Content

OnlineKhabar contains highly dynamic content including:

- Dynamic Wait Handling
- Handling Popups
- Handling Stale Elements
- Real-time news updates
- AJAX-loaded articles
- Advertisement injections
- Lazy-loaded images
- Trending content refreshes

8. User Actions Performed

The automation performs multiple advanced actions.

8.1 Mouse Hover

```
from selenium.WebDriver import ActionChains
```

```
Actions = ActionChains(driver)
```

```
Actions.move_to_element(menu).perform()
```

8.2 Double Click

```
Actions.double_click(element).perform()
```

8.3 Drag and Drop

```
Actions.drag_and_drop(source, target).perform()
```

8.4 Keyboard Actions

```
From selenium.webdriver.common.keys import keys
```

```
Search.send_keys(keys.ENTER)
```

8.5 Scrolling

```
driver.execute_script ("window.scrollTo(0, document.body.scrollHeight;")
```

The objective of this automation project is to:

- Validate critical functionalities of the news portal
- Automate user workflows
- Test responsiveness and performance

- Handle dynamic elements
- Generate reusable scalable automation architecture
- Product defect reports
- Implement maintainable pytest + POM framework

9. Scope of Automation

Modules Covered

Module	Description
Homepage	Verify page loads correctly
Navigation Menu	Validate category navigation
Search Feature	Verify article search
Dynamic News Feed	Validate live content loading
Advertisement Sections	Detect rendering and overlaps
Video/Media Section	Mobile and desktop responsiveness
Footer Links	Validate all important links
Performance	Load testing simulation
Error Handling	Broken pages and invalid searches

10. Browser Compatibility Matrix

Browser	Version	OS	Status
Google Chrome	Latest	Windows 11	PASS
Firefox	Latest	Windows 11	PASS
Microsoft Edge	Latest	Windows 11	PASS

10.1 Device Compatibility Matrix

Device	Resolution	Status
Desktop	1920*1080	PASS
Laptop	1366*768	PASS

11. Test Cases:

11.1 Test Execution Process

- Launch target browser
- Navigate to OnlineKhabar URL

- Execute predefined test cases
- Record results (pass/fail)
- Capture screenshots for defects
- Log issues in bug tracking tool

11.2 Test case table:

Test Case table for Onlinekhabar

12. Implementation:

12.1 Install Dependencies

Pip install selenium pytest pytest-xdist pytest-html

12.2 Driver Factory

From selenium import webdriver

Def get_driver(browser):

 If browser == "chrome":

 Return webdriver.Chrome()

Elif browser == "Firefox":

 Return webdriver.Firefox()

Else:

 Raise Exception ("Unsupported browser")

12.3 Sample Test Case

From selenium import webdriver

From pages.login_page import LoginPage

Import pytest

@pytest.fixture

Def driver():

Def driver():

 Driver = webdriver.Chrome()

 Driver.maximize_window()

 Yield driver

 Driver.quit()

@pytest.mark.parametrize("username,password",[

```

        ("abinashmalla","Omabinash100"),
        ("omabinash","omabinash500")
    ])

    Def test_login_page(driver,username,password):
        Login_page = LoginPage(driver)
        Login.page.open_url(https://www.onlinekhabar.com/login)
        Driver.maximize_window()
        Login_page.enter_username(username)
        Login.page.enter_password(password)
        Login_page.click_login()

        Try:
            If "markets" in driver.page_source:
                Print("Login successful")
                Assert True
            Else:
                Raise AssertionError("Login Failed")

        Except Exception as e:
            Print (f'Error: {str(e)}')
            Assert False

```

13. Bug Reports Identified

Bug 1 – Popup Ad Overlay

Severity

High

Description

Advertisement popup blocks article interaction.

Expected Result

User should access content without obstruction

Actual Result

Popup overlaps article section

Status

Open

Community users have raised similar complaints regarding instructive advertisements.

Bug 2 – Slow Image Loading

Severity

Medium

Description

Article images load slowly on repeated navigation.

Status

Open

Bug 3-Layout Shift During Loading

Severity

Medium

Description

Page content shifts while ads dynamically render.

Status

In Progress

Bug Reports: [Bug Report](#) / [Bug_2nd](#)

14. Risk Analysis

Risks and Mitigation

Risk	Probability	Impact	Mitigation
Dynamic XPath failure	High	High	Use robust locators
Website layout changes	High	Medium	Maintain locator repository
CAPTCHA implementation	Medium	High	Use test environment
CDN blocking automation	Medium	High	Use rotating headers
Slow internet	Medium	Medium	Retry logic
Third-party ad failures	High	Low	Ignore external scripts
Browser updates	Medium	Medium	Cross-browser validation

Memory leaks during scroll	High	High	Add cleanup strategy
AJAX synchronization	High	High	Explicit waits

15. Entry Criteria

- Website should be accessible.
- Browser drivers installed.
- Test environment configured.
- Test data available

15.1 Exit Criteria

- All critical tests executed.
- Major defects reported.
- Reports generated.
- Automation scripts stable.

16. Test Reporting

Use PyTest HTML plugin:

Pip install pytest-html

Pytest --html=reports/report.html

17. Results & Outcomes

Final Execution Summary

Metric	Result
Total Test Cases	25
Passed	22
Failed	3
Pass Percentage	88%
Automation Coverage	90%

18. Key Achievements

- Successfully automated major workflows
- Implemented reusable POM architecture
- Integrated pytest framework
- Managed dynamic content efficiently
- Simulated real user behavior
- Identified UI/performance issues
- Created scalable automation framework

- Generated professional reports

19. Conclusion

The automation testing project for OnlineKhabar successfully demonstrated a complete enterprise-level QA automation implementation using Selenium, Pytest and POM architecture.

The framework handled:

- Dynamic web content
- User interaction automation
- Performance validation
- Bug reporting
- Regression-ready architecture

This solution can be extended further for:

- API testing
- Mobile automation
- AI-based visual testing
- Cloud execution
- Cross-browser grid execution

20. References

- OnlineKhabar Official Website
- OnlineKhabar English Edition
- Selenium Documentation
- Pytest Documentation
- JMeter Documentation

21. Screenshots:

report.html

Report generated on 25-May-2026 at 04:55:26 by pytest.html v4.2.0

Environment

Python	3.14.2
Platform	Windows-11-10.0.26200-SP0
Packages	• pytest: 9.0.3 • pluggy: 1.6.0
Plugins	• html: 4.2.0 • metadata: 3.1.1
JAVA_HOME	C:\Program Files\Java\jdk-25

Summary

13 tests took 00:10:13

(Un)check the boxes to filter the results.

☐ 0 Failed, ☒ 13 Passed, ☐ 0 Skipped, ☐ 0 Expected failures, ☐ 0 Unexpected passes, ☐ 0 Errors, ☐ 0 Reruns, ☐ 0 Retried,

[Show all details](#) / [Hide all details](#)

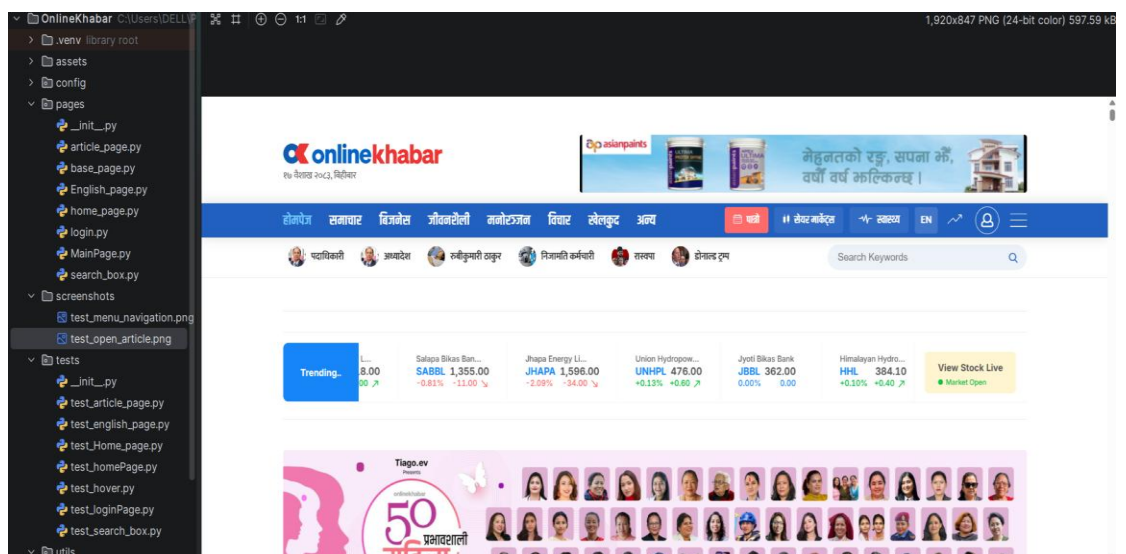
Result	Test	Duration	Links
Passed	tests/test_Login.py::test_login_page[abinashmalla-Ominash100]	00:00:29	
Passed	tests/test_Login.py::test_login_page[omabinash-omabinash500]	00:00:18	
Passed	tests/test_crossBrowsers.py::test_login[chrome-abinashmalla-Ominash100]	00:00:18	
Passed	tests/test_crossBrowsers.py::test_login[chrome-omabinash-omabinash500]	00:00:19	
Passed	tests/test_crossBrowsers.py::test_login[firefox-abinashmalla-Ominash100]	00:00:25	
Passed	tests/test_crossBrowsers.py::test_login[firefox-omabinash-omabinash500]	00:00:25	
Passed	tests/test_crossBrowsers.py::test_login[ChromiumEdge-abinashmalla-Ominash100]	00:00:42	
Passed	tests/test_crossBrowsers.py::test_login[ChromiumEdge-omabinash-omabinash500]	00:00:19	
Passed	tests/test_footer.py::test_footer[chrome]	00:01:23	
Passed	tests/test_links.py::test_links[chrome]	00:00:52	
Passed	tests/test_onlinekhavar_completeworkflow.py::test_workflow	00:03:21	
Passed	tests/test_search_box.py::test_login[chrome]	00:00:46	
Passed	tests/test_search_box.py::test_search	00:00:37	

```
Terminal Local x + v

(.venv) PS C:\Users\DELL\PycharmProjects\News-Portel-Automation> pytest --html=report.html
===== test session starts =====
platform win32 -- Python 3.14.2, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\DELL\PycharmProjects\News-Portel-Automation
plugins: html-4.2.0, metadata-3.1.1
collected 13 items

tests/test_Login.py ..                                     [ 15%]
tests/test_crossBrowsers.py ....
DevTools listening on ws://127.0.0.1:61487/devtools/browser/ab015025-3e13-412b-9e79-648dc2fbf45f
DevTools listening on ws://127.0.0.1:50733/devtools/browser/8d2b0bb1-68f8-4a60-a391-6780a6558287
.                                                         [ 61%]
tests/test_footer.py .                                   [ 69%]
tests/test_links.py .                                    [ 76%]
tests/test_onlinekhavar_completeworkflow.py .           [ 84%]
tests/test_search_box.py ..                             [100%]

----- Generated html report: file:///C:/Users/DELL/PycharmProjects/News-Portel-Automation/report.html -----
===== 13 passed in 613.09s (0:10:13) =====
```



View Results Tree

Name:

Comments:

- Write results to file / Read from file -

Filename

Search: ☐ Case sensitive ☐ Regular exp.

Text

☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request
☒ HTTP Request

Thread Name: users 1-1
Sample Start: 2026-05-07 07:12:01 PDT
Load time: 1206
Connect Time: 933
Latency: 984
Size in bytes: 431915
Sent bytes: 142
Headers size in bytes: 1065
Body size in bytes: 430850
Sample Count: 1
Error Count: 0
Data type ("text"|"bin"|""): text
Response code: 200
Response message: OK

HTTPSampleResult fields:
ContentType: text/html; charset=UTF-8
DataEncoding: UTF-8

View Results in Table

Name:

Comments:

- Write results to file / Read from file -

Filename Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	07:12:01.084	users 1-1	HTTP Request	1206	☒	431915	142	984	933
2	07:13:41.771	users 1-4	HTTP Request	6613	☒	431911	142	190	28
3	07:13:41.836	users 1-6	HTTP Request	7416	☒	431921	142	469	215
4	07:13:41.979	users 1-11	HTTP Request	7546	☒	431911	142	794	329
5	07:13:41.681	users 1-1	HTTP Request	8394	☒	431911	142	241	52
6	07:13:41.711	users 1-2	HTTP Request	8798	☒	431913	142	241	37
7	07:13:41.740	users 1-3	HTTP Request	8805	☒	431913	142	201	24
8	07:13:41.859	users 1-7	HTTP Request	8974	☒	431917	142	483	206
9	07:13:41.836	users 1-5	HTTP Request	9202	☒	431917	142	509	229
10	07:13:42.009	users 1-12	HTTP Request	9961	☒	431913	142	766	297
11	07:13:42.160	users 1-17	HTTP Request	11098	☒	431910	142	923	299
12	07:13:43.389	users 1-58	HTTP Request	11392	☒	431906	142	3070	1952
13	07:13:43.898	users 1-75	HTTP Request	12021	☒	431906	142	3300	1848
14	07:13:44.021	users 1-79	HTTP Request	12086	☒	431906	142	3214	1739
15	07:13:44.078	users 1-81	HTTP Request	12030	☒	431904	142	3157	1686
16	07:13:43.928	users 1-76	HTTP Request	12185	☒	431908	142	3328	1832
17	07:13:43.960	users 1-77	HTTP Request	12167	☒	431908	142	3321	1805
18	07:13:44.230	users 1-86	HTTP Request	11900	☒	431906	142	3315	1565
19	07:13:44.141	users 1-83	HTTP Request	11996	☒	431910	142	3404	1654
20	07:13:43.868	users 1-74	HTTP Request	12516	☒	431906	142	3347	1877
21	07:13:44.109	users 1-82	HTTP Request	12499	☒	431912	142	3401	1685
22	07:13:44.318	users 1-89	HTTP Request	12537	☒	431914	142	3474	1499
23	07:13:44.379	users 1-91	HTTP Request	12476	☒	431908	142	3421	1438
24	07:13:44.260	users 1-87	HTTP Request	12664	☒	431910	142	3597	1557
25	07:13:41.900	users 1-8	HTTP Request	15030	☒	431911	142	474	168

☐ Scroll automatically? ☐ Child samples? No. of Samples: 101 Latest Sample: 20360 Average: 17364 Deviation: 5214

Online Khabar

Timer Group

HTTP Request

HTTP Request

HTTP Request

View Results in Table

Constant Timer

Thread Group

HTTP Request

View Results in Table

View Results Tree

Response Assertion

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
1	05:25:59.299	Thread Grou...	HTTP Request	1237	✓	441758	122	960	847
2	05:25:58.699	Thread Grou...	HTTP Request	1989	✓	441756	122	1561	1447
3	05:25:59.199	Thread Grou...	HTTP Request	1517	✓	441762	122	1060	947
4	05:25:58.800	Thread Grou...	HTTP Request	1940	✓	441756	122	1459	1346
5	05:25:58.899	Thread Grou...	HTTP Request	1868	✓	441760	122	1404	1247
6	05:25:58.999	Thread Grou...	HTTP Request	1824	✓	441756	122	1260	1147
7	05:25:59.098	Thread Grou...	HTTP Request	1732	✓	441762	122	1161	1048
8	05:25:59.399	Thread Grou...	HTTP Request	1440	✓	441746	122	860	747
9	05:25:58.693	Thread Grou...	HTTP Request	2148	✓	441760	122	1566	1453
10	05:25:59.500	Thread Grou...	HTTP Request	1352	✓	441743	122	772	646

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Search:

☐ Case sensitive

☐ Regular exp.

Search

Reset

Text

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

✓ HTTP Request

Sampler result

Request

Response data

Thread Name:Thread Group 2-6

Sample Start:2026-05-25 05:25:59 PDT

Load time:1732

Connect Time:1048

Latency:1161

Size in bytes:441762

Sent bytes:122

Headers size in bytes:1043

Body size in bytes:440719

Sample Count:1

Error Count:0

Data type ("text"|"bin"|""):text

Response code:200

Response message:OK

HTTPSampleResult fields:

ContentType: text/html; charset=UTF-8

DataEncoding: UTF-8